FURTHER MECHANICS / KINEMATICS & DYNAMICS

Impulse and Work Done by a Variable Force

Questions by topic 11 questions · 123 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*7 marks*	2
Question 2	*7 marks*	3
Question 3	*6 marks*	4
Question 4	*8 marks*	5
Question 5	*10 marks*	6
Question 6	*10 marks*	7
Question 7	*11 marks*	8
Question 8	*13 marks*	9
Question 9	*15 marks*	10
Question 10	*16 marks*	11
Question 11	*20 marks*	12

1. A puck of mass 0.40 kg moves on a smooth horizontal surface.
The horizontal unit vectors **i** and **j** are perpendicular.

A hockey stick strikes the puck.
Immediately before contact, the puck has velocity $9\mathbf{i}\,\mathrm{m\,s^{-1}}$.
Immediately after contact, its velocity is $12\mathbf{j}\,\mathrm{m\,s^{-1}}$.

(a) Find the impulse exerted by the stick on the puck, giving your answer as a vector, and calculate its magnitude. **[4]**

(b) During contact, the force exerted by the stick acts in a constant horizontal direction.
Its magnitude, F newtons, is modelled by

$$F = k\sin(25\pi t), \qquad 0 \leq t \leq 0.040$$

where k is a positive constant and t is the time in seconds measured from the start of contact.
The stick loses contact with the puck at $t = 0.040$.
Using your answer to part (a), find the exact value of k. **[3]**

2. A trolley of mass 6 kg moves along a smooth horizontal track at $4\,\mathrm{m\,s^{-1}}$ and strikes a buffer.
The buffer brings the trolley to rest without causing it to rebound.
Air resistance is negligible.

(a) Find the magnitude of the impulse exerted by the buffer on the trolley. **[2]**

(b) The trolley first touches the buffer at time $t = 0$ and comes to rest at time $t = T$ seconds.
During this interval, the buffer exerts a horizontal force opposite to the trolley's motion.
The magnitude of this force is

$$F(t) = 18\,000t(T - t)\,\mathrm{N}, \qquad 0 \leq t \leq T$$

Using your answer to part (a), find the stopping time T and the maximum magnitude of the force exerted by the buffer. **[5]**

3. A sled of mass 20 kg slides down a smooth straight slope inclined at 30° to the horizontal.
It reaches a hydraulic buffer with speed $3\,\mathrm{m\,s^{-1}}$.

The buffer exerts a force on the sled directed up the slope.
When the buffer has been compressed by x metres, the magnitude of this force is F newtons, where

$$F = \frac{A}{1+2x} + 200x$$

and A is a positive constant.
The sled comes to rest after compressing the buffer by 0.5 m.
Throughout the compression, the sled moves along the slope.
Model the sled as a particle.

(a) Find, in terms of A, the work done against the buffer during this compression. **[3]**

(b) Using work and energy, find the exact value of A, taking account of the work done by gravity. **[3]**

4. A trolley of mass 80 kg travels along a straight horizontal track.
A powered launcher exerts a variable driving force on the trolley in the direction of motion for 2 s.
Throughout this interval, a constant resistance of 40 N acts against the motion.

At time $t = 0$, when the launcher begins to act, the trolley has speed $3\,\mathrm{m\,s^{-1}}$.
At time $t = 2\,\mathrm{s}$, its speed is $7\,\mathrm{m\,s^{-1}}$.
The trolley does not change its direction of motion.

(a) Find the magnitude of the impulse exerted on the trolley by the launcher during this interval. **[3]**

(b) The driving force exerted by the launcher is modelled by

$$F = kt^2(2 - t), \qquad 0 \leq t \leq 2$$

where F is measured in newtons, t is measured in seconds and k is a positive constant.

(i) Find the value of k. **[3]**

(ii) Determine the maximum driving force predicted by this model. **[2]**

5. A particle P of mass $2\,\text{kg}$ moves along a straight horizontal line.
Take the positive direction along the line to be the direction of its initial motion.

At time $t = 0$, the velocity of P is $3\,\text{m}\,\text{s}^{-1}$.
For $0 \leq t \leq 2$, the only horizontal force acting on P has signed component

$$F(t) = (3t^2 - 12t + 9)\,\text{N}$$

in the positive direction.
The particle continues to move in the positive direction throughout this interval.

(a) Find the impulse of the force over the interval $0 \leq t \leq 2$. **[3]**

(b) Find the speed of P at $t = 2$. **[2]**

(c) Find the work done by the force during the portion of the interval $0 \leq t \leq 2$ for which the force opposes the motion of P. **[5]**

6. A particle P of mass $4\,\text{kg}$ moves along a straight line on a smooth horizontal surface.
At time $t = 0$, its velocity is $4\,\text{m}\,\text{s}^{-1}$ in the positive direction.
The only horizontal force acting on P has component

$$F(t) = (3t^2 - 12t)\,\text{N}$$

in the positive direction, where t is the time in seconds and $t \geq 0$.

(a) Find the magnitude of the impulse exerted by the force between $t = 0$ and $t = 2$. **[3]**

(b) Find the speed of P at $t = 3$, and state whether it is moving in the positive or negative direction. **[3]**

(c) Find the exact time at which P next comes to rest after $t = 2$. **[4]**

7. A particle of mass 4 kg moves along a straight horizontal line.
A fixed positive direction is chosen along the line.
At time t seconds, where $t \geq 0$, the component of the resultant force in this direction is

$$F(t) = 6(t-1)(t-3)\,\text{N}$$

At time $t = 0$, the velocity of the particle is $-1\,\text{m}\,\text{s}^{-1}$.

(a) Find the magnitude of the impulse exerted on the particle between $t = 0$ and $t = 1$. **[3]**

(b) Find the velocity of the particle at $t = 1$. **[2]**

(c) Hence find all times $t > 0$ at which the speed of the particle is $1\,\text{m}\,\text{s}^{-1}$. **[6]**

8. In this question use $g = 9.8\,\mathrm{m\,s^{-2}}$.

A compact load of mass 60 kg is raised vertically by a cable.
When the load is x metres above its initial position, the tension in the cable is P newtons, where

$$P = 498 + \frac{k}{(x+2)^2}$$

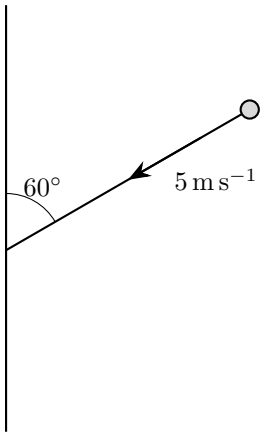
The constant k satisfies $k > 360$.
The load starts from rest at $x = 0$, and the cable remains taut throughout the motion considered.
Air resistance is ignored.

(a) Find, in terms of k and h, the work done by the tension as the load rises from $x = 0$ to $x = h$. Also find, in terms of k, the height at which the tension equals the weight of the load. **[5]**

(b) The load has speed $3\,\mathrm{m\,s^{-1}}$ when $x = 6$. Determine k and find the speed of the load when $x = 8$, giving the speed to three significant figures. **[5]**

(c) Find the height at which the load first comes instantaneously to rest after leaving its initial position. **[2]**

(d) Give one physical reason why neglecting air resistance is reasonable for this load during its ascent. **[1]**



9. A ball of mass 0.4 kg moves on a smooth horizontal surface and collides with a smooth fixed vertical wall. Immediately before impact, its speed is $5\,\mathrm{m\,s^{-1}}$ and its direction makes an angle of 60° with the wall, as shown in the diagram above.

The ball first touches the wall at time $t = 0$ and loses contact at time $t = 0.2\,\mathrm{s}$.
During contact, the magnitude of the force exerted by the wall on the ball is modelled by

$$F = \begin{cases} kt(0.2 - t), & 0 \leq t \leq 0.1 \\ k(0.2 - t)^2, & 0.1 < t \leq 0.2 \end{cases}$$

where F is measured in newtons and k is a positive constant.

(a) Show, by integration, that the magnitude of the impulse exerted by the wall during the collision is

$$\frac{k}{1000}\,\mathrm{N\,s} \qquad \textbf{[5]}$$

(b) The coefficient of restitution between the ball and the wall is known to lie between $\frac{1}{4}$ and $\frac{3}{4}$, inclusive. Determine the corresponding range of values of k, giving exact bounds and fully justifying your answer. **[5]**

(c) Given that $k = 1500\sqrt{3}$, find the speed of the ball immediately after the collision in exact form, and the angle its direction of motion makes with the wall, to the nearest 0.1°. **[5]**

10. A car of mass 800 kg moves along a straight horizontal road.
At time $t = 0$, it passes a marker with speed $20\,\mathrm{m\,s^{-1}}$.
After travelling a distance x m beyond the marker, its speed is $v\,\mathrm{m\,s^{-1}}$.

The driving force varies with distance and has magnitude

$$\left(1800 + 1200e^{-x/200}\right)\ \mathrm{N}$$

The resistance to motion has magnitude $2v^2$ N.

(a) Form a differential equation relating v and x.
By making the substitution $u = v^2$, solve this equation to show that

$$v^2 = 900 + (3x - 500)e^{-x/200}$$ **[8]**

(b) Using the work–energy principle and your result from part (a), find the exact work done against resistance during the first 200 m beyond the marker.
Hence find the average magnitude of the resistance over this distance. **[8]**

11. A particle of mass 2 kg moves along a horizontal straight line.
At time $t = 0$, it passes through a point O with velocity $1\,\mathrm{m\,s^{-1}}$.
The resultant force acts in the direction of motion and has magnitude F N, where

$$F = \frac{27}{v} - 3v$$

and $v\,\mathrm{m\,s^{-1}}$ is the velocity of the particle.
Its displacement from O at time t seconds is x m.

(a) Formulate and solve a differential equation relating v and x to show that

$$x = \ln\left(\frac{3+v}{2(3-v)}\right) - \frac{2}{3}(v-1), \qquad 1 \leq v < 3$$

[6]

(b) Find the exact displacement of the particle when its velocity first reaches $2\,\mathrm{m\,s^{-1}}$.
By changing the variable of integration from x to v, calculate the work done by the force up to this instant.
Hence find the average force over this displacement, giving your answer to three significant figures. **[5]**

(c) Formulate and solve a differential equation relating v and t to obtain an expression for v in terms of t. **[5]**

(d) Calculate the velocities at $t = 1$ and $t = 2$, giving your answers to four significant figures.
Find the impulse of the force over the interval $1 \leq t \leq 2$, giving your answer to three significant figures. **[4]**